

SnoreLess Anti-Snoring Oral Device

MP Judging Presentation

Project ID: 1M21356

11 August 2021



MP Survey Completion

Thank you for completing MP feedback – Matric no: 1906144I



ENGMP <engmp@tp.edu.sg>

8:15 pm



To: ALVIN TEO XI HAO

Dear Student

Thank you for completing the MP Student survey.

Best Regards
ENG MP Admin Support

Thank you for completing MP feedback – Matric no: 1906886E



ENGMP <engmp@tp.edu.sg>

Wed 8/4/2021 12:11 PM

To: CHIN FU JIE



Dear Student

Thank you for completing the MP Student survey.

Best Regards
ENG MP Admin Support

Powered by QuestionPro

[Reply](#) | [Forward](#)

Thank you for completing MP feedback – Matric no: 1900286D



ENGMP <engmp@tp.edu.sg>

Mon 8/2/2021 11:46 AM

To: YANG QUAN KANG, SEAN



Dear Student

Thank you for completing the MP Student survey.

Best Regards
ENG MP Admin Support

Powered by QuestionPro Survey Software

[Reply](#) | [Forward](#)

Our Team



Alvin Teo Xi Hao
(1906144I)



Chin Fu Jie
(1906886E)



Yang Quan Kang, Sean
(1900286D)

Content Overview

1. **Background & Project Information**
2. **Our Solution**
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5. **Commercialisation Potential**
6. **Project Milestones**
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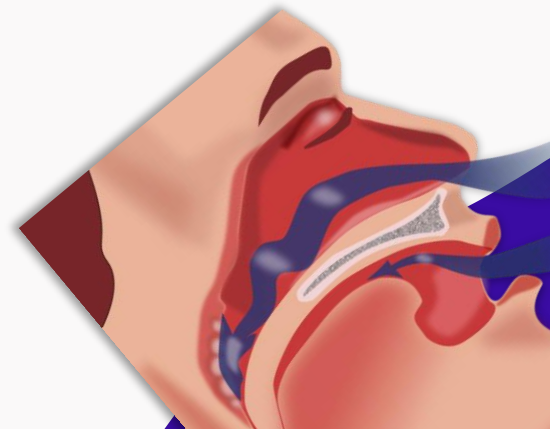
**SnoreLess Anti-Snoring
Oral Device**

BACKGROUND INFORMATION

Background Information



- Snoring (a symptom of sleep apnea) affects quality, comfort of sleep and affect others around them
- Underlying health issues: Chronic diseases, high blood pressure, stroke, diabetes and many more



How Snoring Occurs?

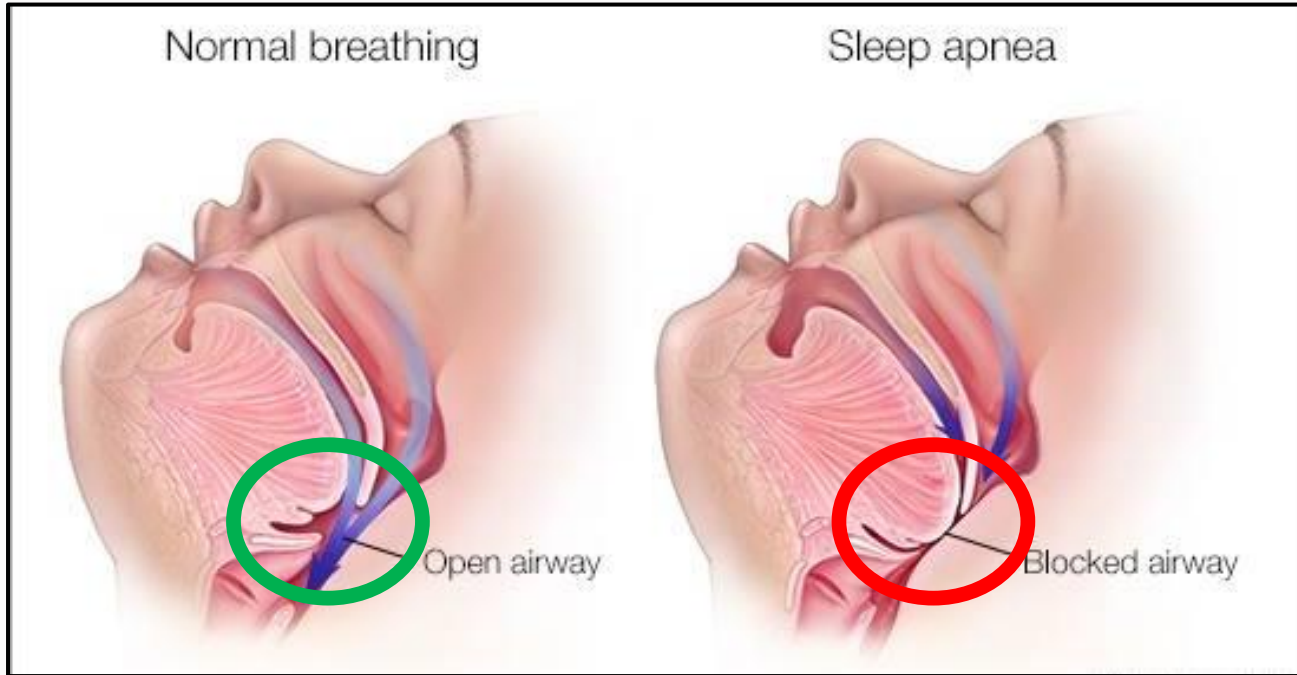
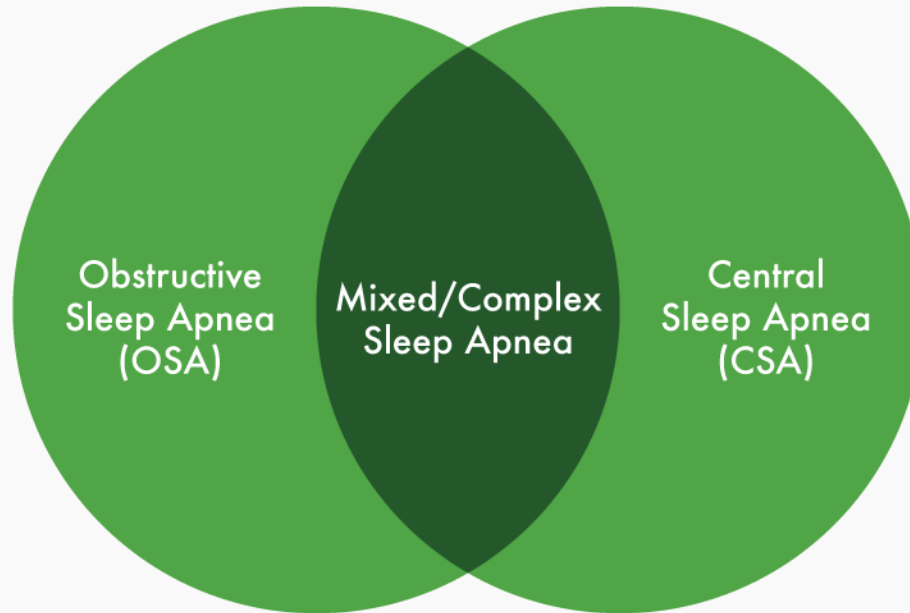


Photo adapted from *Mayo Clinic*

Types of Sleep Apnoea



Statistics

“According to American Academy of Sleep Medicine, about 40 percent of adult men snores frequently and 24 percent of female snores habitually.”

Source: <http://sleepeducation.org/essentials-in-sleep/snoring/overview-and-facts>

**SnoreLess Anti-Snoring
Oral Device**

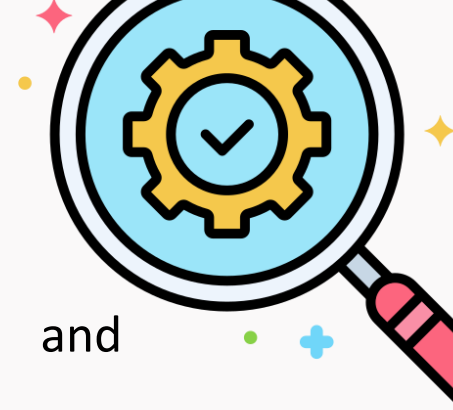
PROJECT DESCRIPTION

Project Objectives



- To develop a low-cost oral medical device that prevents snoring condition and alleviate sleep apnea
- Aims to help the patient to have quality sleep and not to disrupt others with their sleeping conditions
- Project will be submitted for the National Assistive & Rehabilitation Technology Student Innovation Challenge (ARTSIC) competition 2021

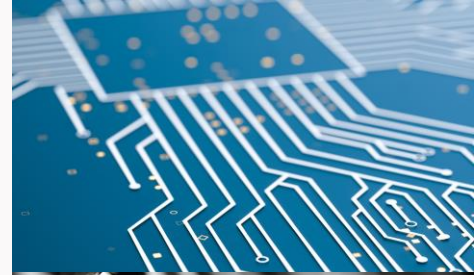
Project Scope



- To conduct survey and research on the causes and interventions of snoring and sleep apnoea
- To learn various machinery and software to aid in the design and production of the oral device
- To develop a functional prototype of an oral medical device that prevents snoring condition and alleviate sleep apnoea

Project Deliverables

- Finalising the 3D CAD model of SnoreLess
- Demonstrating a functional working prototype;
 - i. Fits the mouth
 - ii. Inflatable bag inflates fully
 - iii. Physical tests to show the inflatable tongue retaining mechanism (ITRM) concept clearly



Target Users

- People with long-term snoring conditions
- Patients who want to alleviate sleep apnoea

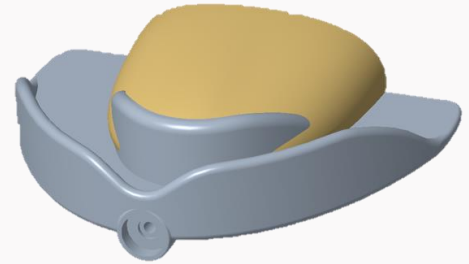


**SnoreLess Anti-Snoring
Oral Device**

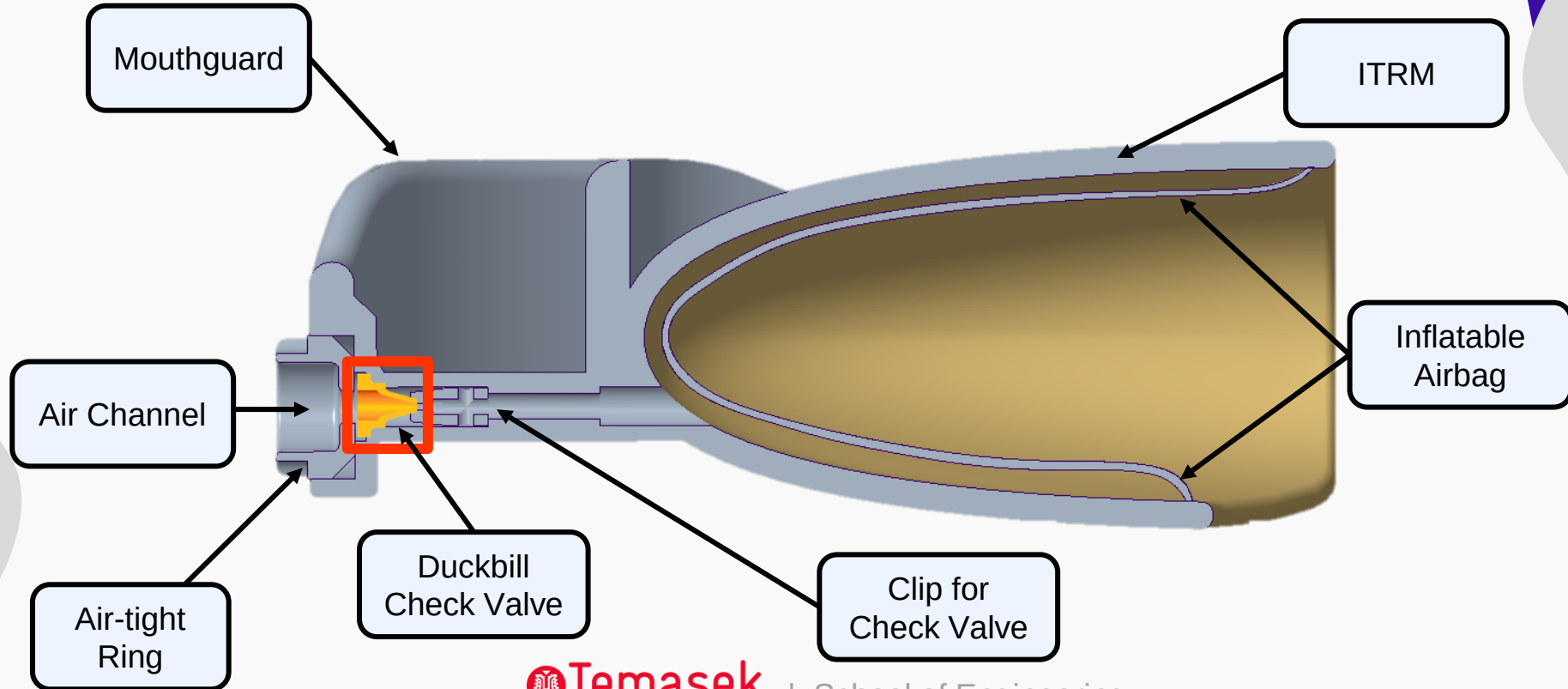
OUR SOLUTION

Working Principles

- Device holds the tongue in place, preventing it from falling back into the windpipe
- Medical Grade Material (MED610)
- Held entirely within the mouth
- Can be used during the day to “train” the tongue
- Conditions and strengthens the tongue – which improves their snoring condition after extensive usage

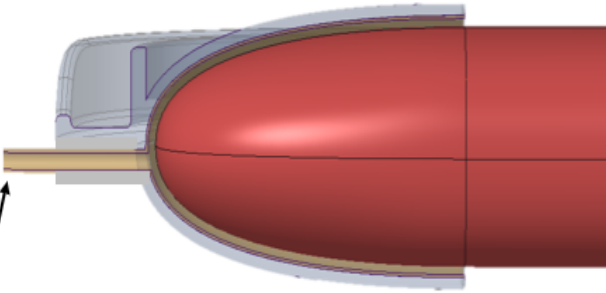


Cross-sectional View of Prototype



How does it work?

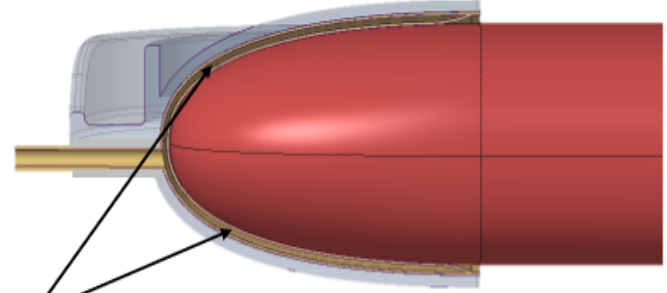
Step 1: User's
tongue engage
the TRD



Air flows
into tube

Before Inflation of Airbag

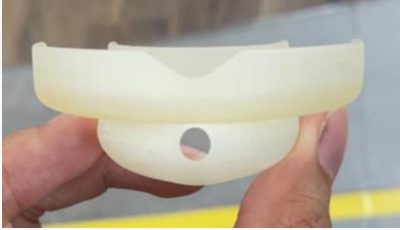
Step 2: Air pumps in through the tube to
inflate airbag



Maintain constant air
pressure to keep
tongue in place

After Inflation of Airbag

Prototype Design Changes



Version 1
(April)

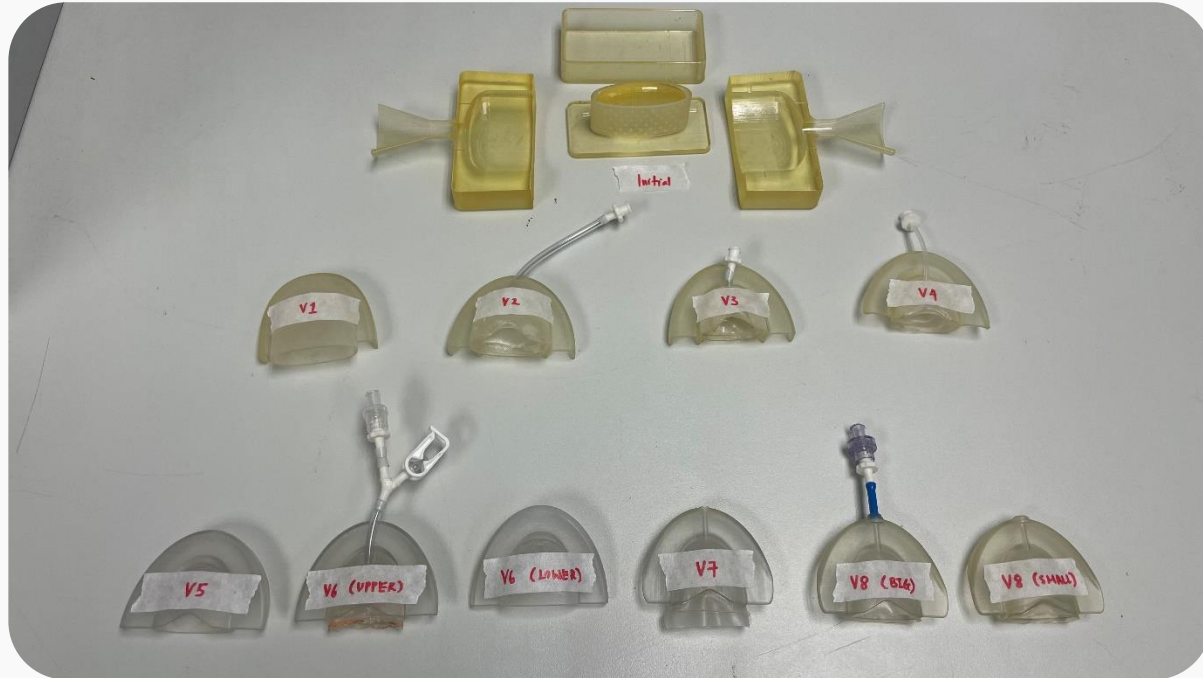


Version 5
(June)

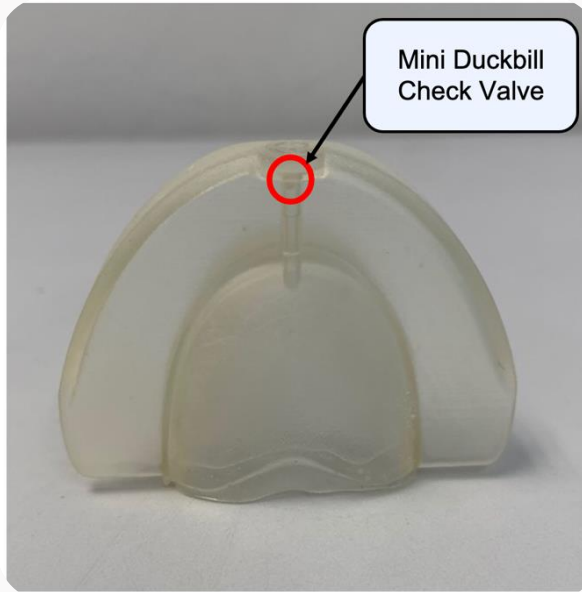


Version 8
(August)

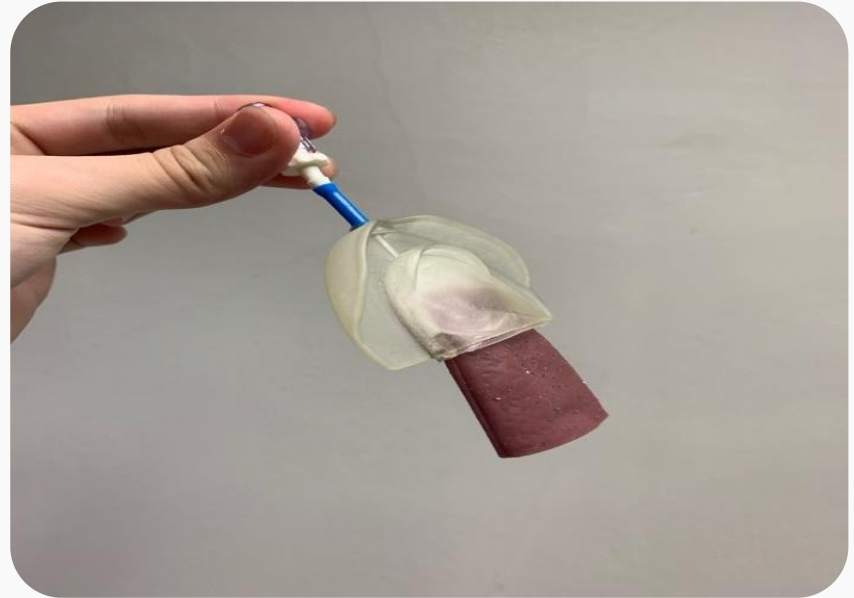
Prototype Design Changes



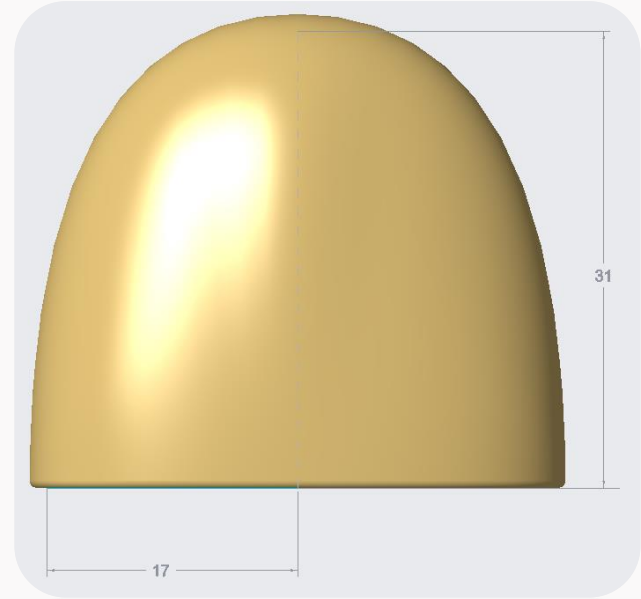
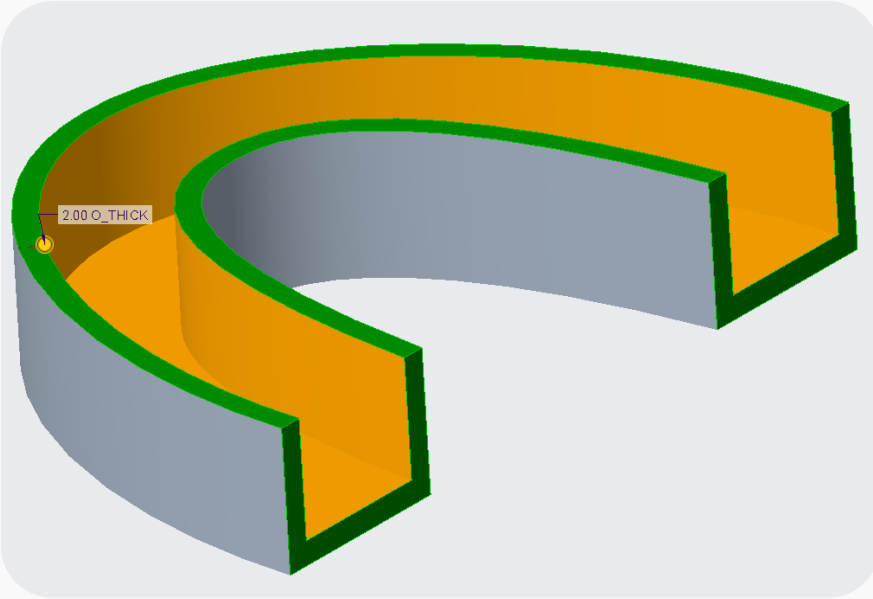
Current Design



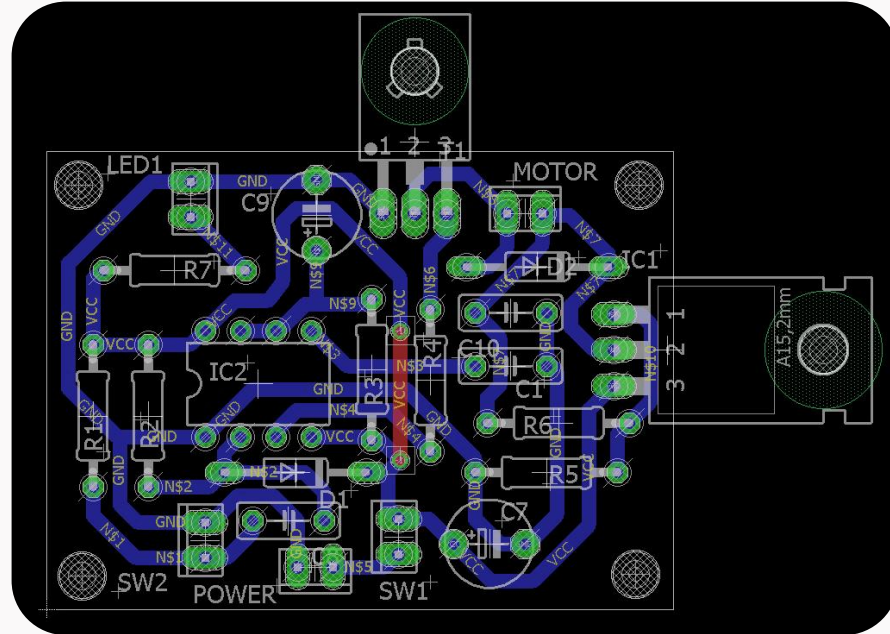
Current Design



Design Measurements

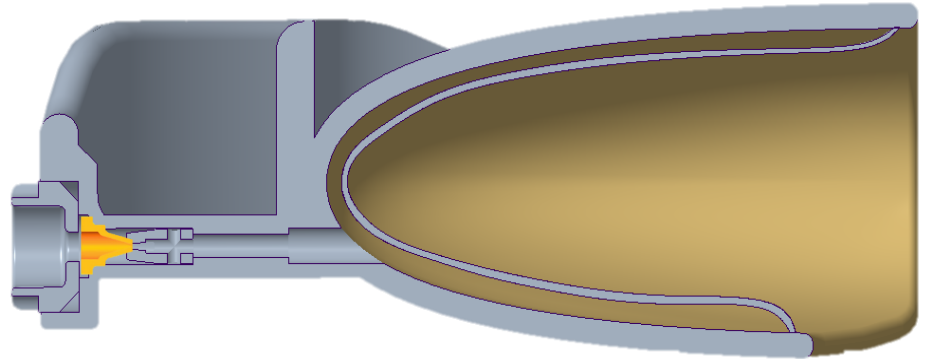


Electronic Air Pump



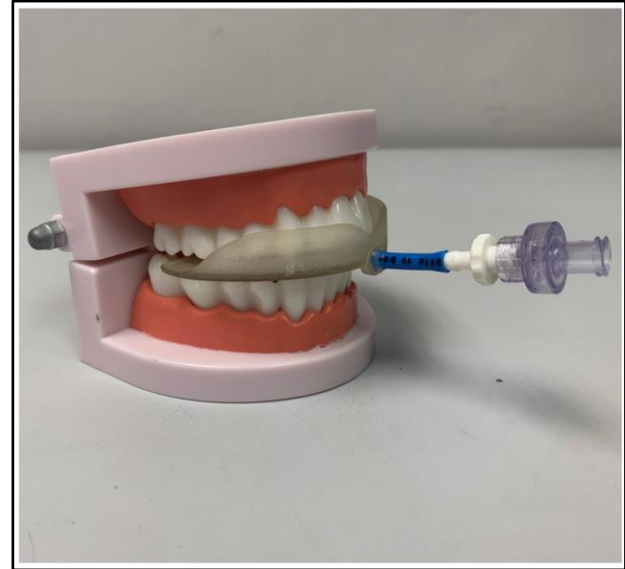
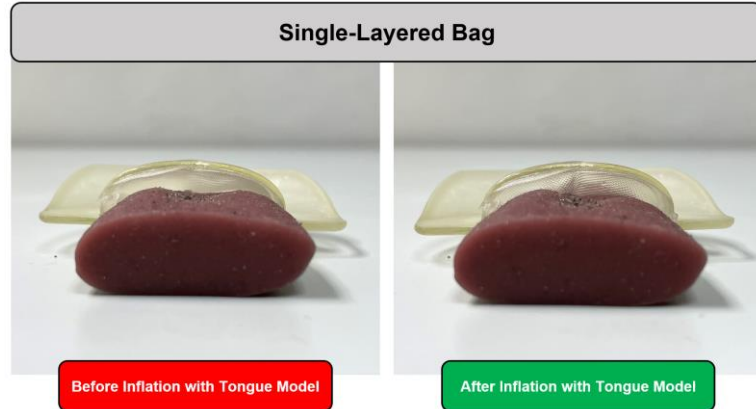
Prototype Demonstration

- Cross Section View



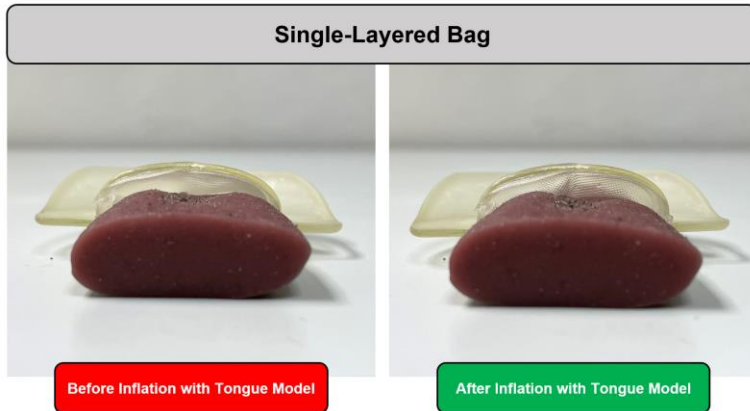
Prototype Demonstration

- Large Check Valve



Prototype Demonstration

- Miniature Duckbill Check Valve



Results

- The mean apnea-hypopnea index decreased significantly from 38 to 14 ($p < 0.001$) with the tongue-retaining device
- May act as an alternative to continuous positive airway pressure



**SnoreLess Anti-Snoring
Oral Device**

BENEFITS & ADVANTAGES

☆ Key Benefits of Our Solution ☆



Prevents snoring and ensures quality sleep



Tongue remains in the mouth in a natural position to ensure comfortability

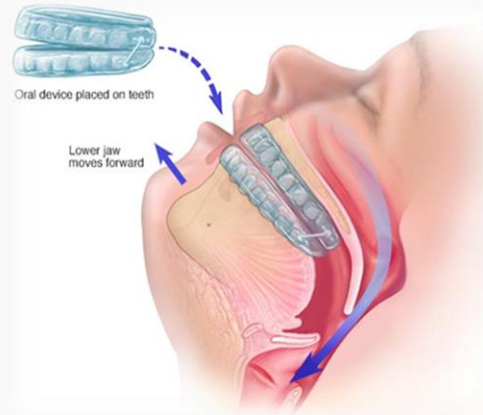


Adaptable and customizable to various sizes of tongue

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MARKET RESEARCH

Current Interventions & Solutions

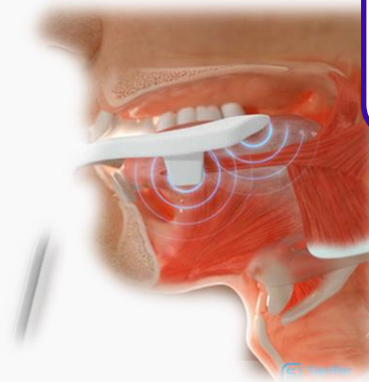


**Mandibular
Advancement
Device (MAD)**

**Tongue Retaining
Device (TRD)**



**Stimulation
Therapy (ST)**



Market Research (Comparison)					
Name	SnoreRX Plus	VitalSleep	Good Morning Snore Solution	SleepTight Mouthpiece	eXciteOSA Sleep Apnea Therapy Device
Type	MAD	MAD	TRD	MAD	ST
Price	SGD 167	SGD 135	SGD 121	SGD 108	SGD 1021

**SnoreLess Anti-Snoring
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MARKET NEEDS ANALYSIS

Anti-Snoring Device Market Survey

via

Google Forms 😊



Anti-Snoring Device Market Survey

Hi everyone,

We are a team of 3 students from the Diploma in Biomedical Engineering (BME) from Temasek Polytechnic embarking on our major project for this semester.

We would like to invite you to help us complete this survey form regarding snoring and anti-snoring device.

This survey would take about 2 minutes of your time to complete.

Thank you very much :)

Cheers,
Project Team

To Access, Click Here:
<https://forms.gle/39gdiqGw7qD5cZXS8>

Next

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Demographics of Respondents

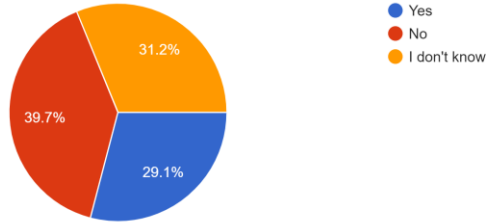
Gender	No. of Responses	Percentage
Male	69	48.9%
Female	72	51.1%

Profile	No. of Responses	Percentage
Student	106	75.2%
Working Adult	31	22.0%
Retiree	2	1.4%
NSF	2	1.4%



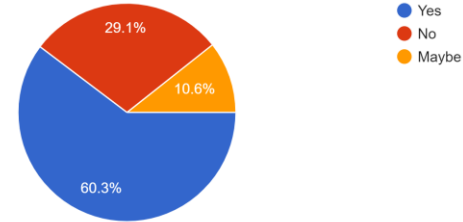
Do you snore when you're asleep

141 responses



Do you get affected by loud snoring sounds from your partner, family members, etc.?

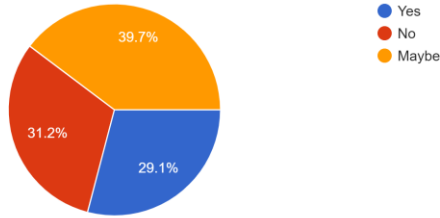
141 responses



- **29.1%** of people snore when they're asleep
- **60.3%** of people have family members that snore

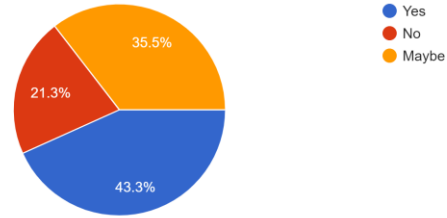
Would you mind having a small device in your mouth if it means you can stop snoring?

141 responses



Would you spend money to acquire an anti-snoring device to help alleviate the sleeping condition?

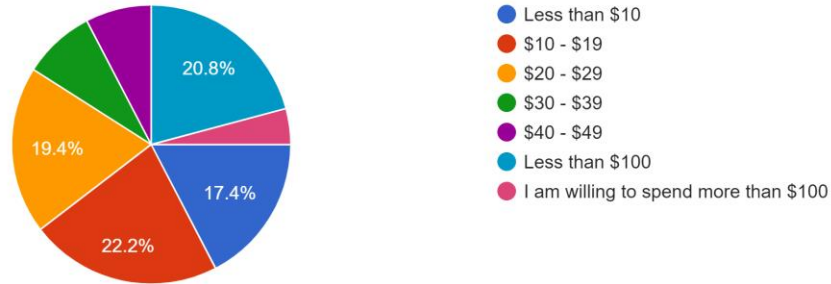
141 responses



- **68.8%** of people are either willing or considering to use a device in their to stop snoring
- **78.8%** of people are either willing or would potentially spend money to buy an anti-snoring device

How much will you be willing to spend?

141 responses



- Generally people would prefer to spend under \$40
- A lot of current devices are typically above the \$40 range
- Estimated cost of device: \$10 – \$20 range

**SnoreLess Anti-Snoring
Oral Device**

COMMERCIALISATION POTENTIAL

Commercialisation Potential



Low-cost device



***Easily
mass produced***



***Customisable
for all***



***SGD \$5.7 billion dollars
market expected to
double in 5 years ^[1]***

^[1]Source: <https://www.gminsights.com/industry-analysis/anti-snoring-treatment-market>

**SnoreLess Anti-Snoring
Oral Device**

PROJECT MILESTONES

Approved Provisional Patent

- Singapore Provisional Patent
- Application No: 10202106620R
- Filing Date: 18 June 2021
- Status: Approved

10202108430T	1) Temasek Polytechnic
10202108056P	1) NATIONAL UNIVERSITY OF SINGAPORE 2) NATIONAL UNIVERSITY HOSPITAL (SINGAPORE) PTE LTD 3) TEMASEK POLYTECHNIC
10202107273X	1) Temasek Polytechnic
10202107285R	1) Temasek Polytechnic
10202107287Q	1) Temasek Polytechnic
10202106620R	1) Temasek Polytechnic
10202102850I	1) Temasek Polytechnic
10202101049U	1) Temasek Polytechnic
10202101155T	1) Temasek Polytechnic



CONFIDENTIAL

INVENTION DISCLOSURE FORM

NOTES TO INVENTORS AND DEVELOPERS

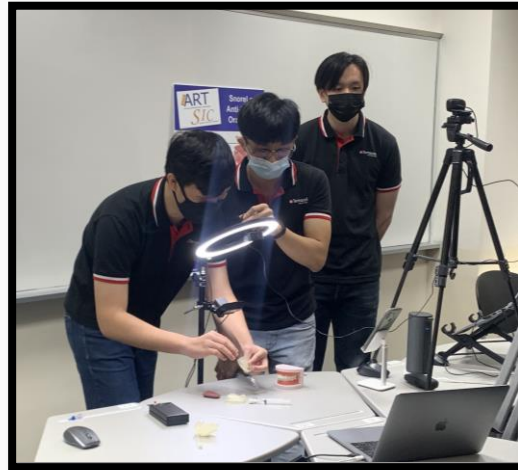
1. The purpose of this form is to inform the Polytechnic of your invention or your student's invention, which may have a positive outlook for commercialization, or pursuant to any agreement with an external party. This form should be submitted when something new and useful has been conceived and developed, or when unusual, unexpected or non-obvious research results have been achieved and can be utilized. The completed form, and any attachments, should be submitted to the Director of Research & Technology Development Department (RTD):
 - through the Director of School/Department, if the sole inventor or one of the inventors is a staff member of the Polytechnic; or
 - through the project supervisor and through the Director of School/Department, if the sole inventor is a student of the Polytechnic or all of the inventors are students of the Polytechnic.
2. The invention should be described clearly including the background, the state of arts, technical merits, novelty, usefulness, and application of the invention. Avoid the use of general statements. If you use any unusual terms or ordinals, please explain in an unusual way, do explain them.
3. In the form, boxed areas under each sub-heading are placeholders and not meant to be restrictive or limiting. Please add as much materials as needed under a particular sub-heading and there is no limit to the number of paragraphs, pages, volumes etc. you can add. Drawings are always helpful and do describe all the parts and how the parts work together.
4. All section of the form should be completed. Indicate "NIL" or "NA" ("Not Applicable") where appropriate.
5. All inventors and developers (if applicable) should read, where applicable, the current Polytechnic Intellectual Property Policy deposited on the Staff Portal before completing this form.
6. RTD manages all technology intellectual property generated by Polytechnic staff members and students. Do not hesitate to contact RTD if you need help or clarification by sending an email to inventions@tp.edu.sg.

National ARTSIC 2021

- Category: Technology
- Team: T11 (Temasek Polytechnic)
- Final results will be out on **13 August 2021 (this Friday), 2pm**

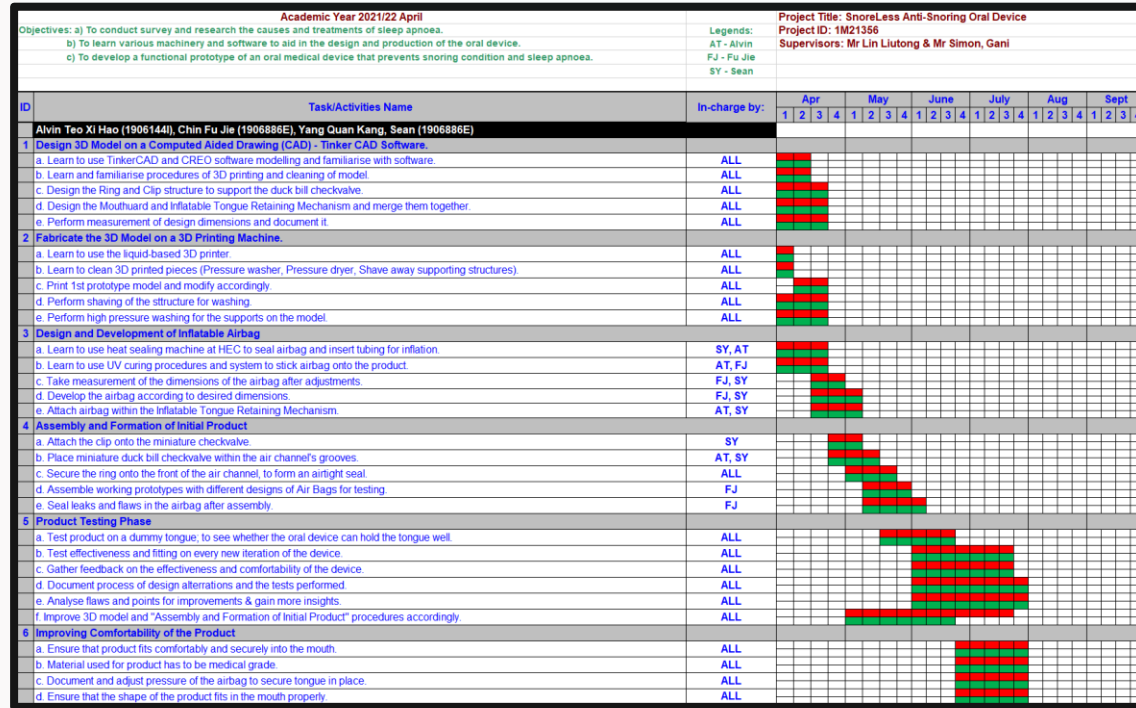


National ARTSIC 2021



SUMMARY

Project Gantt Chart



Future Improvements

Future

Self-inflating Functionality

Flexible Medical Grade Material

Clinical Trials (Possible A*STAR Collaboration or dental clinics)

REFERENCES

References

- [1] Comemorial, “Digital Marketing Background,” Cleanpng.com, n.d. Accessed on: Jun. 17, 2021. [Online]. Available: <https://www.cleanpng.com/png-social-media-marketing-social-networking-service-c-2263198/download-png.html>.
- [2] Freepik, “Day free vector icons designed by Freepik,” Flaticon, 2021. Accessed on: Jun. 17, 2021. [Online]. Available: https://www.flaticon.com/free-icon/day_2206513?term=sleep&page=1&position=6&page=1&position=6&related_id=2206513&origin=tag.
- [3] Imran S. and Seema K., “What is snoring?,” American Academy of Sleep. Nov., 2020. Accessed on: Jun. 10, 2021. [Online]. Available: <https://sleepeducation.org/sleep-disorders/snoring/#what-is-snoring>.
- [4] L. Foley, “Best Anti-Snoring Mouthpieces and Mouthguards,” Sleep Foundation, Jun. 7, 2021. Accessed on: Jun. 17, 2021. [Online]. Available: <https://www.sleepfoundation.org/best-anti-snoring-mouthpieces-and-mouthguards>.

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- [5] Lazada, “Bokali New silicone Anti-Snoring,” n.d. Accessed on Jun. 17, 2021. [Online]. Available: <https://www.lazada.com.my/products/bokali-new-silicone-anti-snoring-tongue->.

- [6] Portsmouth Regional Hospital, “The surprising (and serious) links between sleep, snoring, and stroke,” May 13, 2020. Accessed on: Jun. 3, 2021. [Online]. Available: <https://www.ucsfhealth.org/education/snoring>

- [7] UCSF Health, “Snoring,” n.d. Accessed on: May 30, 2021. [Online]. Available: <https://www.ucsfhealth.org/education/snoring>.

- [8] Wikipedia, “Sleep apnea,” n.d. Accessed on: May 16, 2021. [Online]. Available: https://en.wikipedia.org/wiki/Sleep_apnea.

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Oral Device**

Q & A

THANK YOU!